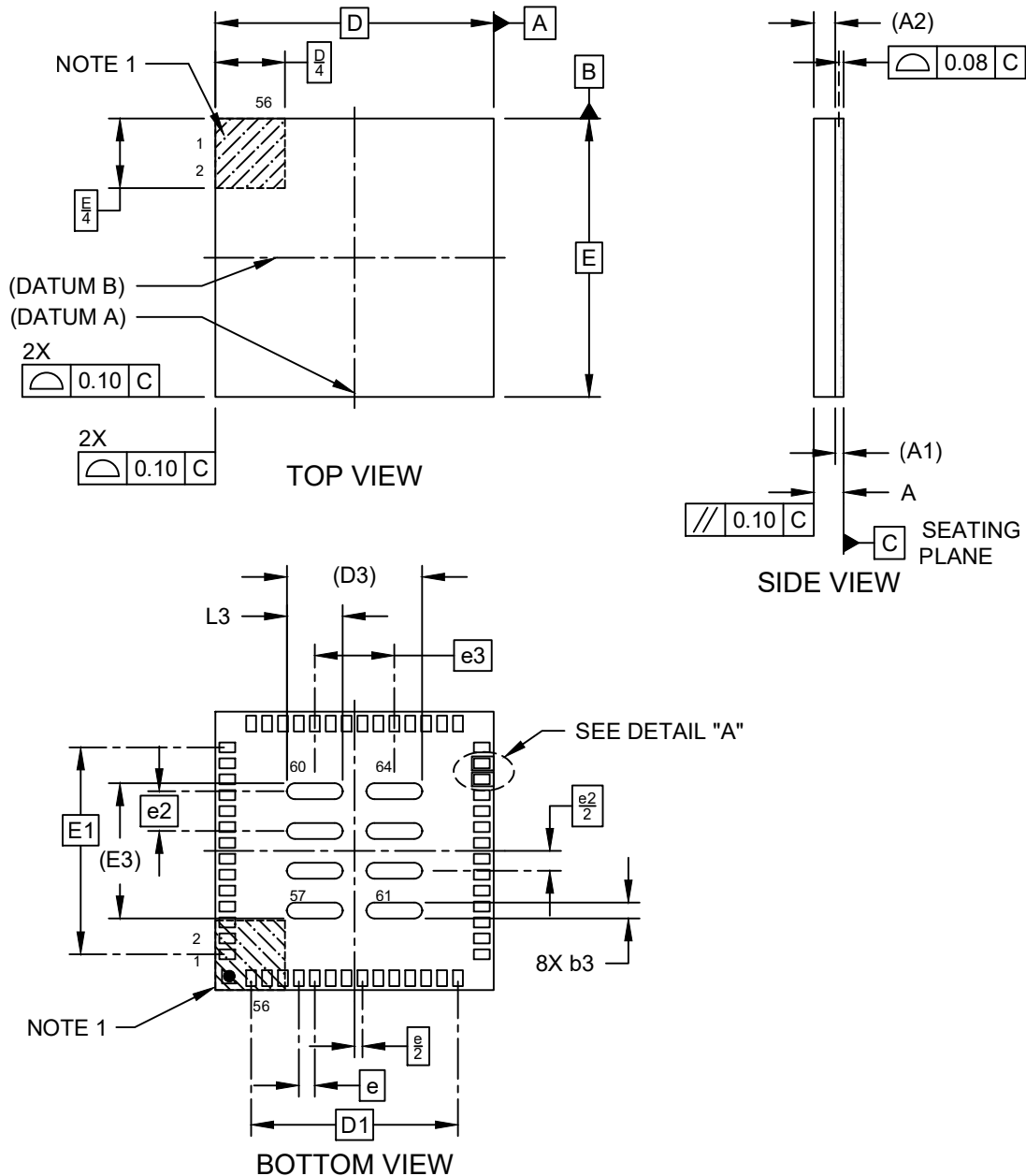


Packaging Diagrams and Parameters

56-Lead Very Thin Fine Pitch Land Grid Array (4W) - 7x7x0.9 mm Body [VFLGA] With Exposed Pads

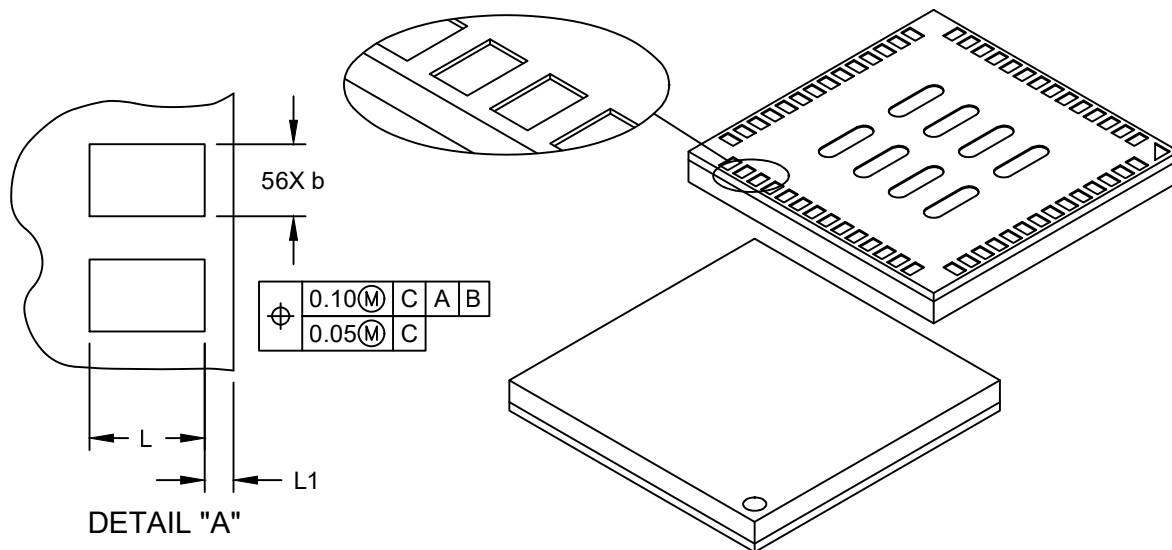
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Packaging Diagrams and Parameters

56-Lead Very Thin Fine Pitch Land Grid Array (4W) - 7x7x0.9 mm Body [VFLGA] With Exposed Pads

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	56		
Pitch	e	0.40 BSC		
Exposed Pads Pitch	e2	1.00 BSC		
Exposed Pads Pitch	e3	2.00 BSC		
Overall Height	A	-	-	0.90
Substrate Thickness	A1	0.21 REF		
Mold Cap Height	A2	0.54 REF		
Overall Length	D	7.00 BSC		
Overall Terminal Pitch	D1	5.20 BSC		
Exposed Pads Overall Length	D3	3.40 REF		
Overall Width	E	7.00 BSC		
Overall Terminal Pitch	E1	5.20 BSC		
Overall Terminal Pitch	E3	3.40 REF		
Terminal Width	b	0.20	0.25	0.30
Exposed Pad Width	b3	0.35	0.40	0.45
Terminal Length	L	0.35	0.40	0.45
Terminal Pullback	L1	-	0.10	-
Exposed Pad Length	L3	1.35	1.40	1.45

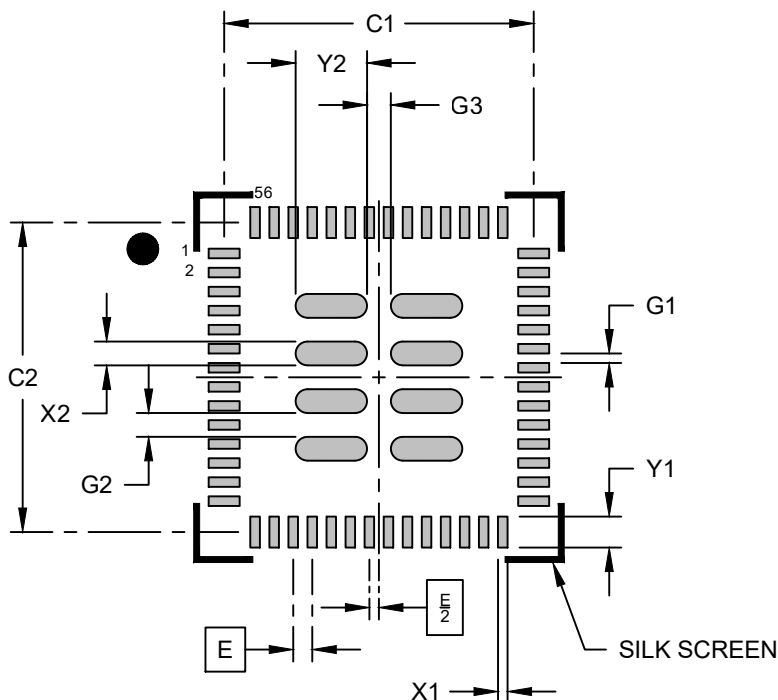
Notes:

1. The Pin 1 visual index feature may vary, but it must be located within the hatched area.
2. The package is saw singulated.
3. Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerances, for information purposes only.

Land Pattern

56-Lead Very Thin Fine Pitch Land Grid Array (4W) - 7x7x0.9 mm Body [VFLGA] With Exposed Pads

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.40 BSC		
Center Pad Width (X8)	X2			0.50
Center Pad Length (X8)	Y2			1.50
Contact Pad Spacing	C1		6.50	
Contact Pad Spacing	C2		6.50	
Contact Pad Width (X56)	X1			0.20
Contact Pad Length (X56)	Y1			0.65
Contact Pad to Pad (X52)	G1	0.20		
Center Pads Clearance (X6)	G2		0.50	
Center Pads Clearance (X4)	G3		0.50	

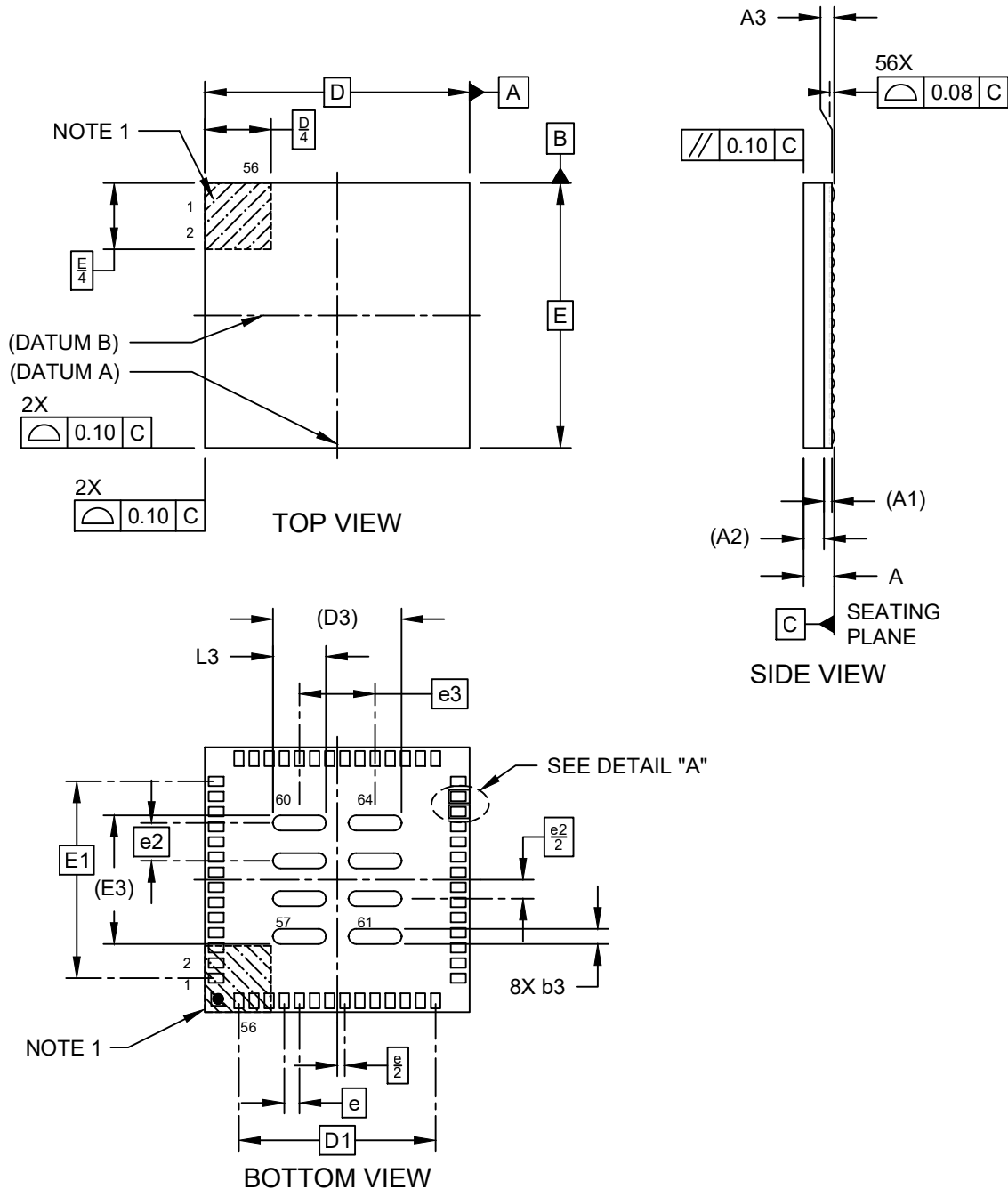
Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during the reflow process.

Packaging Diagrams and Parameters

56-Lead Very Thin Fine Pitch Land Grid Array (8W) - 7x7x1.0 mm Body [VFLGA] With Exposed Pads; Pre-Solder Alternative

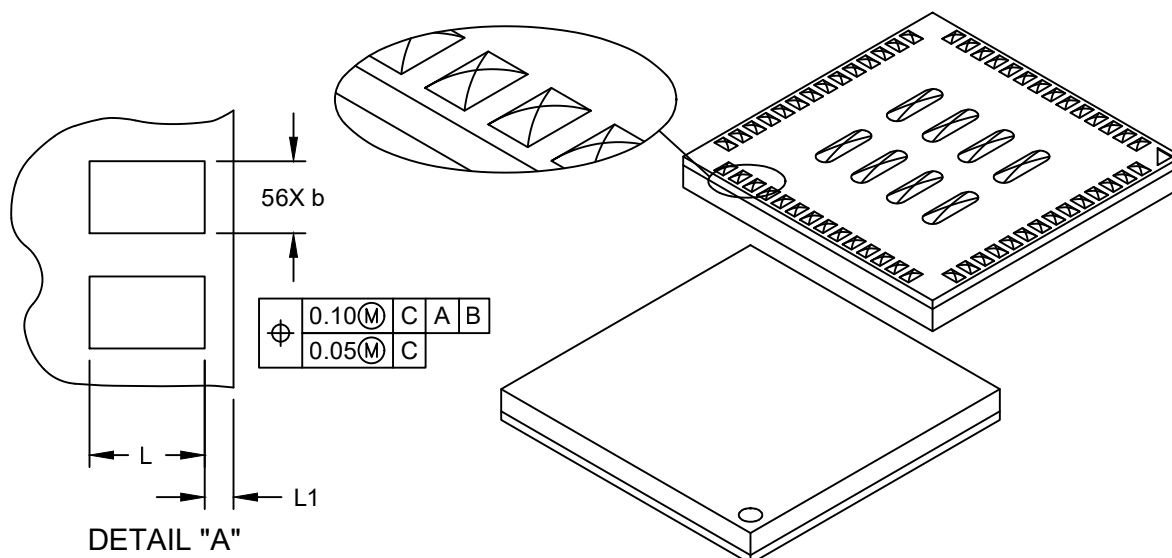
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Packaging Diagrams and Parameters

56-Lead Very Thin Fine Pitch Land Grid Array (8W) - 7x7x1.0 mm Body [VFLGA] With Exposed Pads; Pre-Solder Alternative

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Terminals	N		56		
Pitch	e		0.40 BSC		
Exposed Pads Pitch	e2		1.00 BSC		
Exposed Pads Pitch	e3		2.00 BSC		
Overall Height	A	-	-	-	1.00
Substrate Thickness	A1		0.21 REF		
Mold Cap Height	A2		0.54 REF		
Pre-Solder Height	A3	0.02	0.06	0.10	
Overall Length	D		7.00 BSC		
Overall Terminal Pitch	D1		5.20 BSC		
Exposed Pads Overall Length	D3		3.40 REF		
Overall Width	E		7.00 BSC		
Overall Terminal Pitch	E1		5.20 BSC		
Exposed Pads Overall Width	E3		3.40 REF		
Terminal Width	b	0.20	0.25	0.30	
Exposed Pad Width	b3	0.35	0.40	0.45	
Terminal Length	L	0.35	0.40	0.45	
Terminal Pullback	L1	-	0.10	-	
Exposed Pad Length	L3	1.35	1.40	1.45	

Notes:

1. The Pin 1 visual index feature may vary, but it must be located within the hatched area.
2. The package is saw singulated.
3. Dimensioning and tolerancing per ASME Y14.5M

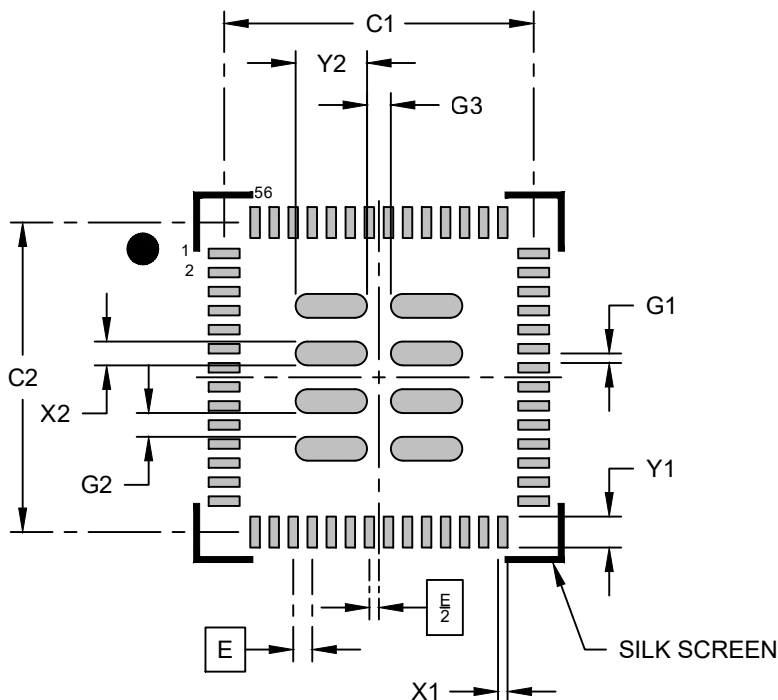
BSC: Basic Dimension. Theoretically exact value shown without tolerances.

REF: Reference Dimension, usually without tolerances, for information purposes only.

Land Pattern

56-Lead Very Thin Fine Pitch Land Grid Array (8W) - 7x7x1.0 mm Body [VFLGA] With Exposed Pads; Pre-Solder Alternative

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.40 BSC		
Center Pad Width (X8)	X2			0.50
Center Pad Length (X8)	Y2			1.50
Contact Pad Spacing	C1		6.50	
Contact Pad Spacing	C2		6.50	
Contact Pad Width (X56)	X1			0.20
Contact Pad Length (X56)	Y1			0.65
Contact Pad to Pad (X52)	G1	0.20		
Center Pads Clearance (X6)	G2		0.50	
Center Pads Clearance (X4)	G3		0.50	

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during the reflow process.